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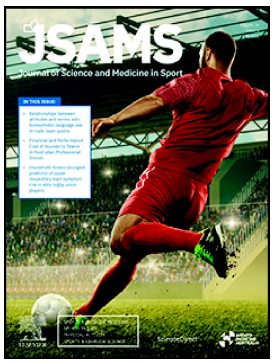

The association between hip/groin pain and hip strength in football players: An exploratory analysis of the FORCe cohort

C. Bonello, M.G. King, K.M. Crossley, J.J. Heerey, M.J. Scholes, P. Lawrenson, M.A. Girdwood, J.L. Kemp, A.B. Mosler, B.F. Mentiplay, A.I. Semciw

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The association between hip/groin pain and hip strength in football players: An exploratory analysis of the FORCe cohort

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Bonello C – this author has had substantial contributions to the design of the work, acquisition of data for the work, drafting and revising the work critically for important intellectual content, provided final approval of the version to be published and agrees to be accountable for all aspects of the work in ensuring that questions related to the accuracy or integrity of any part of the work are appropriately investigated and resolved.

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The association between hip/groin pain and hip strength in football players: An exploratory analysis of the FORCe cohort

Abstract:

Objectives: Explore associations between peak hip strength in football players with hip/groin pain and healthy controls.

Design: Cross-sectional study

Methods: Male and female sub-elite football players (soccer and Australian Football) with hip/groin pain >6-months duration and players without hip/groin pain were recruited across Melbourne and Brisbane, Australia. Demographic information and two questionnaires, the Copenhagen Hip and Groin Outcome Score (HAGOS) and the International Hip Outcome Tool 33 (iHOT) were collected. Hand-held dynamometry was used to measure isometric hip strength for flexion, extension, abduction, adduction, internal rotation, and external rotation. Linear mixed effects models were used to compare strength measures between groups.

Results: 190 football players with hip/groin pain (mean±SD age, 27.8±6.3 years) and 64 controls (age, 27.3±5.6 years) were included in this study. Of these, 291 symptomatic limbs and 128 control limbs were used for analyses. Symptomatic players had lower peak hip adduction(adjusted mean difference= -0.18: 95%CI -0.27 to -0.08, P=0.001), external rotation(-0.06: 95%CI -0.09 to -0.02, P=0.003), and internal rotation strength(-0.06: 95%CI -0.10 to -0.03, P=0.001) compared to controls. A sport-specific interaction was observed for hip abduction strength. When separated by football code, abduction strength was lower in symptomatic Australian football players compared to their same sport peers(-0.20: 95%CI -0.33 to -0.06, P=0.004), but not in symptomatic soccer players(-0.05: 95%CI -0.15 to 0.06, P=0.382).

Conclusions: Hip adduction, internal rotation, and external rotation strength appear lower in football players with hip/groin pain independent of sex and football code. Hip abduction strength was lower in symptomatic Australian football players but not in soccer players.

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Keywords: hip, groin, rehabilitation, muscle strength, soccer, Australian Football

Introduction

Hip and groin pain (hip/groin pain) is common amongst active young adults, and symptoms are often longstanding. In professional male Football (soccer), hip/groin injuries contribute to 14-20% of time-loss injuries[1] and 10% in sub-elite male soccer.[2] Injury surveillance in elite Australian Football demonstrates an incidence of 2.28 injuries per club over a season.[3] Given the persistent nature of symptoms, time-loss definitions may significantly underestimate the true burden and impact of injury. Time-loss measures only capture 10% of groin problems in male amateur soccer[4], and 31% of sub-elite players report pain lasting greater than six weeks.[5] In sub-elite female soccer, the overall prevalence of hip/groin injury is 27%, with one in four players reporting a non-time-loss hip/groin injury.[6] Hip/groin pain not only negatively impacts sporting performance but also affects the players' daily function and participation in vocational and leisure activities.[5]

Hip muscle weakness and atrophy have been associated with a variety of hip/groin pain conditions,[7] including adductor-related groin pain,[8] femoroacetabular impingement (FAI) syndrome[9] and chondral pathology.[10] As such, hip muscle strength testing is often recommended during the physical examination of patients with hip/groin pain.[7] Extensive research has highlighted that strength impairments exist in football players with groin pain.[7] Specifically, players with groin pain display lower hip adduction/abduction strength ratios and eccentric adduction strength;[11] and hip adductor muscle strength is decreased preceding and during the onset of a groin injury.[12] Reduced adduction strength is consistently found to be a risk factor for groin pain.[11, 13-15] However, less information about football players with hip/groin pain from a potential hip joint origin (i.e. FAI syndrome or dysplasia) is known.

Sporting demands may influence a player's hip/groin pain presentation and the presence of physical impairments. Research in male Gaelic and rugby players with FAI syndrome demonstrated deficits in flexion, abduction, and internal rotation strength compared to controls.[16] Conversely, in male ice-hockey players, hip muscle strength did not significantly differ between players with FAI

syndrome, players with cam morphology and no symptoms, and controls.[17] However, these studies limit their populations to male-only cohorts, a common theme in hip/groin pain research,[18] and thus, it is unclear if such findings apply to women with hip/groin pain or even other sporting codes.

Current recommendations indicate that physiotherapist-led exercise therapy involving hip muscle strengthening should be at least three months in duration and target individual modifiable physical impairments.[19] However, the scarcity of information relating to the effect of an individual's sport of choice and their sex on potential strength impairments renders it difficult to provide recommendations on potential strength impairments aside from male athletes in limited sports. Therefore, this study aimed to describe the association between hip/groin pain and hip muscle strength in football players (soccer and Australian Football); and determine if sex, or sport (soccer/Australian Football) modified this relationship.

Methods

This cross-sectional, secondary analysis study was conducted using baseline data from the Femoroacetabular Impingement and Osteoarthritis Cohort (FORCe) study; a prospective cohort study investigating hip/groin pain in sub-elite Football (soccer and Australian Football).[20] Ethics approval was obtained from the La Trobe University Human Ethics Committee (HEC 15-019, HEC16-045) and the University of Queensland Medical Research Ethics Committee (No: 24 2015000916, 2016001694), and all participants provided informed consent.

Male and female football players with hip/groin pain who met specific criteria and control football players (Supplementary 1), were recruited via social media, mail-outs, and direct communication with football leagues and clubs across Melbourne and Brisbane, Australia.

On the day of testing, participants attended either La Trobe University or the University of Queensland. Demographic data, including: sex, age, height, body mass, dominant kicking limb, and football code were recorded. Participants completed the Copenhagen Hip and Groin Outcome Score[21] and the International Hip Outcome Tool 33.[22] Both questions are valid and recommended for use in young to middle-aged individuals with hip and groin pain.[21, 23] In addition, symptomatic players were

asked to quantify their average level of pain from 0-10 using a numerical pain rating scale (NPRS) and provide an estimation of symptom duration in months.

The JTECH medical Commander muscle tester (Utah, USA)™ was used for all strength tests. Isometric "make" hip muscle strength tests for flexion, extension, abduction, adduction, internal rotation, and external rotation were examined in standardised positions based on previously used, reliable methods.[20]

Each participant performed one sub-maximal familiarisation test for each hip muscle group, followed by three maximum voluntary contractions. Each isometric muscle contraction was performed for five seconds, with a five-second break between trials. A 30-second break was provided prior to testing the next muscle group. The tester instructed the participant when to commence using standardised verbal encouragement: "go ahead and push-push-push-push-relax". The highest of the three maximal voluntary contractions was used for analysis. Intra-rater reliability for our hip muscle strength testing procedures in this cohort were determined with ICCs ranging from 0.74 to 0.96.

It should be noted that there were a number of variations that existed between sites. For example, assessors were unique at each testing site, which had implications on the way strength testing was performed. In Brisbane, participants remained with their arms folded across their chest for hip flexion strength testing, whereas in Melbourne, participants were asked to grip the side of the plinth. These differences, together with other unknown factors may affect the within site variability of strength measures, and for this reason, site was included as a cluster variable within the analysis.

Peak strength (N) was converted to a torque (Nm) and subsequently normalised to participants' body mass (Nm/kg). Torque was calculated by multiplying peak strength by the lever arm of the test (perpendicular distance from the centre of rotation to the position of the dynamometer). Included participants were sub-categorised based on limb inclusion type (Figure 1).

Linear mixed-effects models were used to compare differences in hip muscle strength (dependent variable) between groups (independent variable, hip/groin pain group vs controls) . This relationship was adjusted for the potential confounders of age (years), sex (female/male), sport (soccer/Australian football), and clustered by site (Melbourne/Brisbane). As demands on hip function differs between football codes, sporting code exposure may influence outcome variable. Participant

ID was included as a random factor to account for the various combinations of limb symptom status per participant (Figure 1), which differs from the original protocol paper which anaylsed participants rather than limbs.[20] Interactions between group (hip/groin pain and controls) and sex, as well as group and sport were investigated. Significant interactions were followed up with stratified analyses (stratified by sex or sport). All analyses were performed using R© version 4.1.1 (https://www.r-project.org/). Linear mixed-effects models were performed through the lme4 package (version 1.1-27.1).

Results

Demographic characteristics for football players with hip/groin pain and control football players are provided in Table 1. A total of 191 football players with hip/groin pain and 69 control football players consented to participate in the study. Strength measures were unable to be recorded for one symptomatic player (male, 27 years old, soccer) and two control players (both male, 25 years old, Australian Football) and limb length was not recorded for an additional three control players (male, 27 years old, soccer; male 22 years old, Australian Football; male 32 years old Australian Football). Rendering 190 participants in the symptomatic group and 64 in the control group. In addition, extension strength was unable to be recorded for one symptomatic player (male, 28 years old, soccer), and flexion strength was unable to be recorded for one control player (male, 27 years old, soccer). When participants were sub-categorised by limb, a total of 291 limbs met the symptomatic inclusion criteria, and 128 control limbs satisfied the control group criteria (Figure 1).

There was insufficient evidence of sex-specific interactions for any strength test (flexion, P=0.551; extension, P=0.138; abduction, P=0.208; adduction, P=0.073; external rotation, P=0.060; internal rotation, P=0.051). There was evidence to support a sport-specific interaction for hip abduction strength (P=0.027), but not for any other strength test (flexion, P=0.648; extension, P=0.584; adduction, P=0.108; external rotation, P=0.086; internal rotation, P=0.131).

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There was insufficient evidence to suggest an association between peak flexion torque and the presence of symptoms after controlling for sex, age, sport, and site (estimate 0.07. 95% CI: -0.15 to 0.01, P=0.103) [Figure 2A & Supplementary 2].

There was insufficient evidence to suggest an association between peak extension torque and the presence of symptoms after controlling for sex, age, sport, and site (estimate -0.07. 95% CI: -0.18 to 0.05, P=0.249) [Figure 2B & Supplementary 2].

There is evidence to suggest that external rotation torque (estimate -0.06. 95% CI: -0.09 to -0.02, P 0.003) and internal rotation torque (estimate -0.06. 95% CI: -0.10 to -0.03, P=0.001) is lower in symptomatic limbs, controlling for sex, age, sport, and site. [Figure 2C-D & Supplementary 2].

The relationship between hip abduction torque and group was modified by the sport played. In Australian football players, there is evidence to support a relationship between weaker hip abduction strength of symptomatic limbs compared to controls (estimate -0.20. 95% CI: -0.33 to -0.06, P=0.004) [Figure 2E & Supplementary 2], when controlling for sex, age, and site. . However, in soccer players, there is insufficient evidence to support a relationship between peak abduction torque and symptoms (estimate -0.05. 95% CI: -0.15 to 0.06, P 0.382) [Figure 2F & Supplementary 2].

There is evidence to support a relationship between lower peak adduction torque for symptomatic limbs compared with control limbs (estimate -0.18. 95% CI: -0.27 to -0.08, P=0.001), controlling for sex, age, sport, and site [Figure 2G & Supplementary 2].

Discussion

This study explored the associations between hip strength in football players with and without hip/groin pain. Overall, those with hip/groin pain were weaker in adduction, internal, and external rotation. These results were neither sex nor sport-specific. A sport-specific effect was observed in abduction strength, where abduction strength was lower in Australian football players with hip/groin pain compared to controls, but this was not observed in soccer players.

Consistent with previous literature, our cohort demonstrated lower strength in hip adduction, and internal and external rotation compared to controls.[7] It may be hypothesised that reductions in hip rotation and adduction strength during change of direction and cutting tasks may negatively influence pelvic stability, force dispersion and possibly contribute to the persistent nature of the pain.[24] Inversely, hip/groin pain may play a role in inhibiting muscle contractions around the hip.
As study participants were currently training and playing at least 2 sessions per week, strength impairments observed were not the result of time-loss based deconditioning. It is more likely that strength of the hip adductors and rotators (internal and external) may have been reduced due to pain or reflex mechanisms.[25] Symptomatic players in this cohort did not display weaker hip flexion strength when compared to controls, contrasting findings from previous studies in pre-operative hip patients.[9, 10] As football players in our study continued full sport participation; ongoing forceful kicking and sprinting movements may explain the maintenance of sagittal plane strength. In Australian Football, the psoas muscle on the kicking leg is shown to have a significantly greater cross-sectional area and supports the maintenance of hip flexion strength.[26] Similarly, hip extension strength is preserved likely due to repetitive sagittal plane movements such as running and jumping.

Adduction weakness was the largest effect observed in symptomatic players and may be an important deficit to address through rehabilitation. Reduced hip adduction strength is commonly reported in athletes presenting with hip/groin pain.[27] Clinically, pain and reduced strength on the adductor squeeze test are frequently used to differentiate athletes with hip/groin pain from those without.[13] Eccentric hip adduction strength was lower in soccer players with adductor-related groin pain,[11] and a 28% adduction strength deficit was present in FAI syndrome.[28] Furthermore, lower peak isometric adductor strength was a significant predictor of future time-loss injuries in men's professional soccer.[8, 14] Implementing progressive adductor exercises improved eccentric hip adduction and abduction strength in professional footballers and has shown to be effective for groin injury prevention in sub-elite players.[29] The consistency of the relationship with previous research, together with the time-course of weakness effecting pain,[8, 14] provides strong evidence for addressing adduction strength deficits to minimise the risk of future hip/groin pain.

Australian football players with hip/groin pain presented with lower hip abduction strength than their same sport peers; a finding not observed in our soccer players. Reduced abduction strength may have clinical implications as the abductor muscles optimise the position of the pelvis relative to the femur.[25] This is critical to prevent movement into a position that impinges the hip (flexion, adduction, internal rotation) during single leg weight-bearing tasks, such as those commonly required in sport. Femoral external rotation and abduction is observed during ball control movements and initiation of the passing action in soccer as well as change of direction skills. The repetitive nature of this movement throughout training and competition may preserve strength in these muscles. This may explain why abduction weakness was only observed in Australian football players (where kicking is a sagittal plane movement). Ipsilateral hip adduction/abduction strength ratio is used as a guideline for evaluating recovery in soccer players with groin problems[15] and in symptomatic Australian football players, strengthening programs may focus on addressing deficits present in hip abduction as well as adduction. While we do present some evidence to support a modifying effect of sport on the relationship between symptoms and strength, we must also acknowledge that 6 interaction tests were performed to observe such an effect. There is still a possibility that our outcomes reflect a Type I error.

Exercise-based rehabilitation programs for football players with longstanding hip/groin pain may target strength deficits observed in this study. High-value exercise interventions should focus on addressing the individualised impairments identified in the physical examination and progress towards functional and/or sport-specific rehabilitation.[19] Findings from this study improve our understanding of the nature and extent of strength deficits in football players with hip/groin pain and support the implementation of structured, evidence-based, strengthening programs such as the adductor strengthening program.[29] What cannot be gleaned from the cross-sectional nature of the findings is the temporal relationship between strength and hip/groin pain. However, these findings provide valuable insights into future longitudinal evaluations of strength impairments to determine causality. Subsequent research should evaluate if the listed impairments result in greater hip/groin pain or if hip/groin pain result in strength impairments and the influence sport choice has on these outcomes.

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This study included football players with self-reported history of hip/groin pain and a positive flexion-adduction-internal rotation (FADIR) test. The authors acknowledge the intimate and complex anatomical relationship of the hip and groin region and the difficulty in evaluating the true pathoanatomical source of symptoms. The poor specificity of the FADIR test means that findings cannot be assumed to be solely specific to the hip joint.[30] Secondly, we were unable to recruit an even distribution of male and female football players (in the symptomatic or control groups). Thus, we may have been underpowered to determine the presents of any sex specific effects which requires consideration when interpreting the findings. Further, through the exclusion of planned surgery, previous significant hip conditions and contraindications, this may introduce selection bias. Our study benefits from a large cohort of participants, across two codes of Football, utilising a clinical diagnosis of hip/groin pain which adds to the overall generalisability of the results.

Conclusion

Hip adduction, internal rotation, and external rotation strength was lower in footballers with hip/groin pain compared to controls regardless of football code played. Hip abduction strength was lower in Australian football players with hip/groin pain, but not in soccer players. These impairments were present despite players not experiencing a time-loss injury. These findings may guide clinicians in impairments that could be targeted during the physical examination and subsequent strengthening programs for the management and prevention of hip/groin pain.

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Practical Implications:

- Hip adduction, internal rotation, and external rotation strength was lower in soccer and Australian footballers with hip/groin pain compared to controls
- Hip abduction strength was lower in Australian football players with hip/groin pain, but not in soccer players
- Hip flexion and extension strength did not differ between soccer and Australian footballers with hip/groin pain compared to controls

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Table 1: Demographic characteristics for football players with hip/groin pain and control players

Demographics	**Football players with hip/groin pain (n= 190)**	**Control football players (n= 64)**
Age, y	27.8 ± 6.3	27.4 ± 5.8
Sex, male [n (%)] : female [n (%)]	male [152 (80.0%)]: female [38 (20.0%)]	male [47 (73.4%)]: female [17 (26.6%)]
Height, m	1.78 ± 0.08	1.77 ± 0.1
Body Mass, kg	78.9 ± 13.2	76.9 ± 12.6
BMI, kg/m^2	24.7 ± 3.2	24.4 ± 2.8
Dominant limb, Right n (%)	168 (88.4%)	49 (76.6%)
Football Code, Soccer [n (%)]: AF [n (%)]	Soccer [99 (52.1%)]: AF [91 (47.9%)]	Soccer [38 (59.4%)]: AF [26 (40.6%)]
Total competitive sessions per week (training and game)	3.00 ± 0.5	2.91 ± 0.9
Symptom duration (months)	24 [16, 50]	-
Patient-reported outcome measures		
iHOT (total)	62.1 ± 17.5	96.5 ± 6.4
HAGOS		
HAGOS (symptoms)	59.1 ± 14.4	96.2 ± 6.0
HAGOS (pain)	72.7 ± 13.9	99.8 ± 1.0
HAGOS (physical function in daily living)	79.7 ± 16.4	100.0 ± 0.0
HAGOS (function in sport and recreation)	63.4 ± 18.0	98.5 ± 5.4
HAGOS (participation in	61.6 ± 25.1	96.9 ± 7.8

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physical activities)		
HAGOS (quality of life)	59.1 ± 16.4	98.4 ± 6.1

Abbreviations: y, years; m, metres; kg, kilogram; BMI, body mass index; AF, Australian football

Values are presented as % or mean ± standard deviation or median [IQR]

Figure Legends:

Figure 1: Limb coding and number of included limbs based on limb sub-categories

Limbs with hip/groin pain that met the inclusion criteria were included and coded as 1, 2, or 4. All control participant limbs were coded as 6. Hips coded as 3 (hip/groin pain but not meeting study inclusion criteria (Supplementary 1)) were excluded due to the presence of symptoms. Hips coded as 5 were excluded due to potential biomechanical and strength alterations resulting from a contralateral symptomatic hip.

Figure 2: Forest plot illustrating the adjusted mean differences and 95% confidence intervals for all strength variables compared between groups.

Figures:

Figure 1:

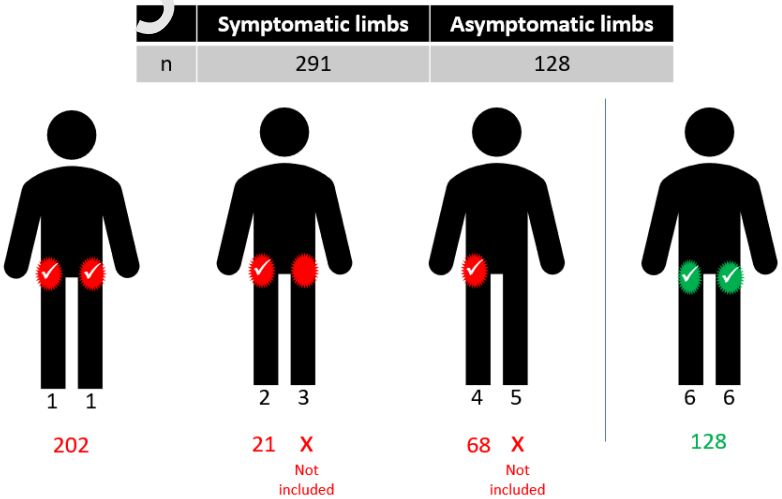
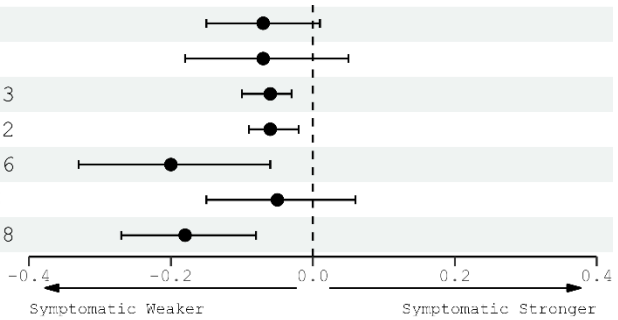


Figure 2:

Strength	Estimate	LowerCI	UpperCI
Flexion	-0.07	-0.15	0.01
Extension	-0.07	-0.18	0.05
Internal Rotation	-0.06	-0.1	-0.03
External Rotation	-0.06	-0.09	-0.02
Abduction (AFL)	-0.2	-0.33	-0.06
Abduction (Soccer)	-0.05	-0.15	0.06
Adduction	-0.18	-0.27	-0.08



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CRediT Author Statement

Christian Bonello: Conceptualization, Writing – original draft, Writing – review and editing, project administration, data curation. **Matthew King:** Conceptualization, Writing – original draft, Writing – review and editing, project administration, investigation, data curation. **Kay Crossley:** funding acquisition, supervision, writing – review and editing. **Joshua Heerey:** investigation, data curation, writing – review and editing. **Mark Scholes:** investigation, data curation, writing – review and editing. **Peter Lawrenson:** investigation, data curation. **Mick Girdwood:** Data curation. **Joanne Kemp:** Funding acquisition, writing – review and editing. **Andrea Mosler:** Writing – review and editing. **Ben Mentiplay:** Data curation, writing – review and editing. **Adam Semciw:** Conceptualization, Writing – original draft, Writing – review and editing, software, validation, formal analysis, investigation, data curation

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